Course Essentials

Summary

- 14 lectures, on Tuesday from 1:15pm to 3pm (one 15-minute break around 2pm).
- Assistant : Luiz Bissoto (luiz.bissoto@epfl.ch).
- Grading based on **3 problem sets**, weights **40%, 40%, 20%**. Assignments can be done in groups of up to 3 people.

Three parts of the course

- Basics of non-cooperative game theory (approximately 7 lectures)
- Auction theory with applications to Ebay and Google advertising auctions (approx. 4 lectures)
- Evolutionary game theory (approx. 2 lectures)

Cooperative vs Non-Cooperative Games

Definition

NON-COOPERATIVE GAME
There is **no external authority enforcing rules of cooperative behavior**.
Each player tries to maximize her benefit at any time, even by damaging others. Even when a contract between two players could be beneficial to both, they cannot stipulate it, because no authority is able to enforce it.

Definition

COOPERATIVE GAME
Involves negotiation, bargaining, contracts,...

Key idea

Classical game theory focuses on **non-cooperative games** : this is what we study in this course.

What is a Game?

Definition

GAME
A mathematical model of strategic interaction, defined by :

- **Players** – people, businesses, countries...
- **Choices** – players make one or multiple choices between different possible actions. A player's **strategy** is her rule for choosing actions.
- **Outcomes** – when players have made their choices, an outcome is realized and the game ends. An outcome is expressed in terms of a **payout** to the players (e.g. Player 1 wins $1, Player 2 loses $2).

Key idea

Common to all games : **ALL PLAYERS CARE ABOUT IS THEIR OWN PAYOUT!!**

Elements distinguishing different types of game

- **Timing** – do players choose once (e.g. rock-paper-scissors) or repeatedly over time (e.g. chess)? Does the game eventually end, or continue forever? Do they choose **simultaneously** or in turn?
- **Observation** – can the players observe each other's choices?

This week : the simplest type of games

- 2 players;
- each player makes one choice;
- all choose simultaneously (so do not observe the other's action).

Normal Form Representation

Definition

NORMAL FORM
A table with one row per action of Player 1, one column per action of Player 2, and in each cell the payouts **(Player 1, Player 2)**.

Example

Rock-Paper-Scissors

$1\backslash 2$	R	P	S
R	(0,0)	(-1,1)	(1,-1)
P	(1,-1)	(0,0)	(-1,1)
S	(-1,1)	(1,-1)	(0,0)

- Players : 1, 2
- Possible actions for Player 1 (and for Player 2) : R(ock), P(aper), S(cissors).

Key formulas

Payout notation :

$$\mathcal{P}^{(1)}(R, P) = -1$$

Payout for Player 1 if 1 chooses R and 2 chooses P.

$$\mathcal{P}^{(2)}(S, R) = 1$$

Payout for Player 2 if 1 chooses S and 2 chooses R.

Pure vs Mixed Strategies

Player i can choose between M **actions** $(a_1^{(i)}, a_2^{(i)}, \ldots, a_M^{(i)})$.

Definition

MIXED STRATEGY
An M-component vector

$$s^{(i)} = (p_1^{(i)}, p_2^{(i)}, \ldots, p_M^{(i)})$$

with $0 \leq p_j^{(i)} \leq 1$ for $j = 1, \ldots, M$ and $\sum_{j=1}^{M} p_j^{(i)} = 1$.
Each component j is the probability of choosing action $a_j^{(i)}$.
Different players choose their actions independently.

Definition

PURE STRATEGY
A vector with one component equal to one, and the others equal to zero.

Actions vs Strategies

Warning

- **Action** (letter a) : what a player ends up playing.
- **Strategy** (letter s) : the M-component vector containing the *ex-ante* probability of playing each of the different actions.

Key idea

When players use mixed strategies, their objective is to maximize their **expected payout**.
Expected value of a random variable : weighted average over possible outcomes, weighed with their respective probabilities.

Key formulas

2 players, each choosing between actions A and B : $s^{(1)} = (p_A^{(1)}, p_B^{(1)})$, $s^{(2)} = (p_A^{(2)}, p_B^{(2)})$.

$$\begin{aligned} E[\mathcal{P}^{(1)}(s^{(1)}, s^{(2)})] = &\, Prob(A,A)\,\mathcal{P}^{(1)}(A,A) \\ &+ Prob(A,B)\,\mathcal{P}^{(1)}(A,B) \\ &+ Prob(B,A)\,\mathcal{P}^{(1)}(B,A) \\ &+ Prob(B,B)\,\mathcal{P}^{(1)}(B,B) \end{aligned}$$

The choices are independent : $Prob(A,A) = p_A^{(1)} p_A^{(2)}$, $Prob(A,B) = p_A^{(1)} p_B^{(2)}$, etc. Hence

$$\begin{aligned} E[\mathcal{P}^{(1)}(s^{(1)}, s^{(2)})] = &\, p_A^{(1)} p_A^{(2)} \mathcal{P}^{(1)}(A,A) \\ &+ p_A^{(1)} p_B^{(2)} \mathcal{P}^{(1)}(A,B) \\ &+ p_B^{(1)} p_A^{(2)} \mathcal{P}^{(1)}(B,A) \\ &+ p_B^{(1)} p_B^{(2)} \mathcal{P}^{(1)}(B,B) \end{aligned}$$

Key idea

In the simultaneous-move games of this week, each player **knows the other players' strategy, but not their actions**.

Example

In Rock-Paper-Scissors, I may know that my opponent plays the three actions with probability $(0.3, 0.3, 0.4)$ – e.g. because we already played together many times – but I don't know what she will play this time.

Nash Equilibrium

Definition

NASH EQUILIBRIUM (NE)
The most widely used **solution concept**.
Given a game with N players, a **strategy profile** (the set of the strategies taken by each of the players) $(s^{(1)}, s^{(2)}, \dots, s^{(N)})$ is a Nash equilibrium iff the payout of each player i is such that

$$\mathcal{P}^{(i)}(s^{(i)}, s^{(-i)}) \geq \mathcal{P}^{(i)}(\tilde{s}^{(i)}, s^{(-i)}) \quad \text{for every possible } \tilde{s}^{(i)}$$

where $s^{(-i)}$ denotes the strategies of all players *other than* i.

Key idea

A NE is the best that one player can do *given the strategies of all the other players*.

Pure-strategy Nash equilibrium

NE obtained when we restrict ourselves to the case in which all players adopt pure strategies (i.e. always play the same action) : a profile in which the action of each player is the **best response** to the action of the other players.

Example

$1\backslash 2$	L	R
U	(8,7)	(4,6)
D	(6,5)	(7,8)

- **Player 1** : if 2 moves Left, 1 should move Up ; if 2 moves Right, 1 should move Down.
- **Player 2** : if 1 moves Up, 2 should move Left ; if 1 moves Down, 2 should move Right.

Circling each best response, the cells where both payouts are circled are the NE :

1 \ 2	L	R
U	(8,7)	(4,6)
D	(6,5)	(7,8)

2 NE : (U, L) and (D, R).

Warning

Game with two equilibria : the theory cannot tell which one will be chosen.

Classical Example 1 – Prisoner Dilemma (PD)

Developed at the RAND corporation in 1950 (organization started immediately after WWII to develop research for military strategy and planning), as one of the first steps in thinking about cold war strategy.

Example

Classic story
Two suspects of a crime are placed in different rooms and interrogated by the police.

- Both **cooperate (C)** with each other and stay silent : 1 year of prison each.
- One **defects (D)** and denounces the other, who stays silent (C) : the one denounced gets 5 years, the denouncer walks away.
- Both defect (D) : 3 years of prison each.

	C	D
C	(-1,-1)	(-5,0)
D	(0,-5)	(-3,-3)

Key idea

- Each player strictly prefers to Defect no matter what the other does (**Defect is a "dominant strategy"**).
- Defect-Defect is the **only Nash Equilibrium**, and it's at the same time the outcome that maximizes social loss!

Definition

PRISONER DILEMMA – GENERAL FORM
Payouts can change, but we call "Prisoner Dilemma" a game of the form

	C	D
C	(X_2, Y_2)	(X_4, Y_1)
D	(X_1, Y_4)	(X_3, Y_3)

with $X_1 > X_2 > X_3 > X_4$ and $Y_1 > Y_2 > Y_3 > Y_4$.

Applications

Example

Coke-Pepsi advertising war

- Without any advertising, each company earns \$5b/year.
- Each company can choose to spend \$2b/year on advertising.
- Advertising does not increase total sales, but if one company advertises while the other doesn't, it captures \$3b/year from the competitor.

	No Ad	Ad
No Ad	(5,5)	(2,6)
Ad	(6,2)	(3,3)

The only NE is (Ad, Ad), with payout (3,3), while (No Ad, No Ad) would give (5,5) to both.

Example

Doping in sports (same matrix : No Dop / Dop)

- Two professional athletes of similar ability compete in 10 tournaments per year, winning on average 5 each.
- If only one of them takes the medication, he wins 8 of the 10 tournaments, leaving only 2 to the opponent. The long-term health cost makes his final utility the same as winning only 6 tournaments and staying healthy.
- If both take it, they both win 5 tournaments each and both pay the health cost (equivalent to 2 tournaments).

Example

Is disarmament possible ? Two "great powers" :

- Is it possible to reach an agreement to dramatically reduce weapons ? Who is going to enforce the agreement if one deviates ?
- Deviating is the best course of action for each player, both to be able to "strike first" and to be able to defend itself.
- Unfortunately, disarmament is not a Nash Equilibrium.

Classical Example 2 – Matching Pennies

Example

Two players choose Head or Tail. Player 1 wins 1 if the two choices match (2 Heads or 2 Tails) and loses 1 otherwise. The other way around for Player 2 : she wins 1 in case of mismatch and loses 1 in case of a match.

	H	T
H	(1,-1)	(-1,1)
T	(-1,1)	(1,-1)

This is a **zero sum game**.

Warning

At first sight, it looks like there is no Nash Equilibrium…But what is true is that **there is no Nash Equilibrium in pure strategy**.

Mixed strategy NE

Player 1 plays H with probability $p_H^{(1)}$ (and T with probability $1 - p_H^{(1)}$). What should Player 2 do ?

Key formulas

If Player 2 plays H, his expected payout is

$$-p_H^{(1)} + (1 - p_H^{(1)}) = 1 - 2p_H^{(1)}$$

If Player 2 plays T, his expected payout is

$$p_H^{(1)} - (1 - p_H^{(1)}) = -1 + 2p_H^{(1)}$$

- If $p_H^{(1)} > \frac{1}{2}$, playing T gives Player 2 a higher expected payout : he should play T with probability 1.
- If $p_H^{(1)} < \frac{1}{2}$, Player 2 should play H with probability 1.
- The only possibility to induce Player 2 to randomize is $p_H^{(1)} = \frac{1}{2}$: to randomize, he must be **indifferent** between the two strategies.

— Same reasoning for the reaction of Player 1 to Player 2's strategy.

Key idea

NE : both players play H and T with probability $\frac{1}{2}$.

Summary

Logic of a mixed-strategy equilibrium

— At least one player plays a mixed strategy, i.e. randomizes between at least two actions.
— To randomize between two actions, a player must be **indifferent** between the two, otherwise she would play the one with higher expected payout.
— For Player 1 to randomize, Player 2 must play H/T with probability 50% each.
— For Player 2 to randomize with 50% each, he must also be indifferent between H and T.
— If both play H/T with probability 50% each, both are indifferent : **both are making an optimal choice** (they could not do anything better) → it is a NE.

Example

Real life : penalty kicks
The goalkeeper has a higher probability of stopping the ball if he jumps on the side where the ball is kicked (Left/Right); the kicker has a higher probability of scoring if he kicks on the opposite side.
Chiappori, Levitt, Groseclose (2002) : statistical study of actual penalty kicks, with three possible actions (Left/Middle/Right) – finds evidence of randomization but also of player heterogeneity (some players prefer to kick on the left, others on the right).
Other applications : tennis, baseball, etc.

Classical Example 3 – Asymmetric Matching Pennies

	H	T
H	(X,-X)	(-1,1)
T	(-1,1)	(1,-1)

X is a big number, such as 1000 or 1 million.

Example

Intuition

— Player 1 might think : I'll play H, just in case Player 2 also plays H : I'll be rich! (and if not, I lose just 1).
— But then Player 2 will never play H, he will always play T! So Player 1 playing H loses 1 for sure.
— So Player 1 should play T and win 1 for sure…Then Player 2 will not play T for sure, because he doesn't want to lose for sure…
— After all, also in this case **both players should randomize**.

Key idea

In a mixed-strategy NE, **each player must be indifferent between playing H and T.**

Key formulas

Player 2 indifferent (given $p_H^{(1)}$) :
— 2 plays H : $-Xp_H^{(1)} + (1-p_H^{(1)}) = 1-(1+X)p_H^{(1)}$
— 2 plays T : $p_H^{(1)} - (1-p_H^{(1)}) = 2p_H^{(1)} - 1$

$$\Rightarrow \quad p_H^{(1)} = \frac{2}{3+X}$$

Player 1 indifferent (given $p_H^{(2)}$) :
— 1 plays H : $Xp_H^{(2)} - (1-p_H^{(2)}) = -1+(1+X)p_H^{(2)}$
— 1 plays T : $-p_H^{(2)} + (1-p_H^{(2)}) = 1-2p_H^{(2)}$

$$\Rightarrow \quad p_H^{(2)} = \frac{2}{3+X}$$

Key idea

NE for

$$s^{(1)} = s^{(2)} = \left(\frac{2}{3+X},\ 1-\frac{2}{3+X}\right)$$

The probability of the big gain for Player 1 (and big loss for Player 2) is $\left(\frac{2}{3+X}\right)^2$.

Classical Example 4 – Split or Steal

	Split	Steal
Split	(1,1)	(0,2)
Steal	(2,0)	(0,0)

Key idea

Main difference with the Prisoner Dilemma : if the opponent steals, I am **indifferent** between stealing or splitting (I get nothing in both cases).
Three pure-strategy Nash equilibria : (Split, Steal), (Steal, Split) and (Steal, Steal).

Which equilibrium would be more plausible?

— If I am uncertain about what the opponent will play, it's better to play Steal.
— If the opponent steals, I am more likely to want to "punish" him.

Example

Split or Steal on British TV
The game is played in a show on British TV (episode watched in class).
One player manages to steer the opponent toward Split, by convincing him that he will play Steal! Why does it work?

— When the opponent knows you will play Steal, he will want to punish you.
— But if he knows that you play Steal and he is rational, he is indifferent between Split and Steal. So even a promise that is not fully credible of splitting the win outside the game can steer him toward Split.

Warning

"Fairness" and "promises" are outside the scope of the theory.

Weekly Summary

Summary

Main concepts

- Cooperative vs non-cooperative games
- Actions vs strategies
- Pure strategies vs mixed strategies
- Normal form representation of a game
- Nash equilibrium

Classical games

- Prisoner Dilemma
- Matching Pennies
- Asymmetric Matching Pennies
- Split or Steal